VII SEMESTER (PROFESSIONAL ELECTIVE-V)	L	T	P	C
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20AM7E07 - SOCIAL NETWORK ANALYSIS				

COURSE OBJECTIVES:

1. Formalize different types of entities and relationships as nodes and edges and represent this information as relational data
2. Plan and execute network analytical computations
3. Use advanced network analysis software to generate visualizations and perform empirical investigations of network data
4. Interpret and synthesize the meaning of the results with respect to a question, goal, or task
5. Collect network data in different ways and from different sources while adhering to legal standards and ethics standards

COURSE OUTCOMES:

After the completion of the course the students are able to,

CO1: Know basic notation and terminology used in network science
CO2: Be able to visualize, summarize and compare networks
CO3: Illustrate basic principles behind network analysis algorithms
CO4: Develop practical skills of network analysis in R programming language
CO5: Be capable of analyzing real work networks

UNIT -I:

Social Network Analysis: Preliminaries and definitions, Erdos Number Project, Centrality measures, Balance and Homophily.

UNIT -II:

Random graph models: Random graphs and alternative models, Models of network growth, Navigation in social Networks, Cohesive subgroups, Multidimensional Scaling, Structural equivalence, roles and positions.

UNIT -III:

Network topology and diffusion, Contagion in Networks, Complex contagion, Percolation and information, Navigation in Networks Revisited.

UNIT -IV:

Small world experiments, small world models, origins of small world, Heavy tails, Small Diameter, Clustering of connectivity, The Erdos Renyi Model, Clustering Models.

UNIT -V:

Network structure -Important vertices and page rank algorithm, towards rational dynamics in networks, basics of game theory, Coloring and consensus, biased voting, network formation games, network structure and equilibrium, behavioral experiments, Spatial and agent-based models.

Text Books:

1. S. Wasserman and K. Faust. “Social Network Analysis: Methods and Applications”, Cambridge University Press.
2. D. Easley and J. Kleinberg, “Networks, Crowds and Markets: Reasoning about a highly connected world” , Cambridge University Press, 1st edition,2010

Reference Books:

1. Maarten van Steen. “Graph Theory and Complex Networks. An Introduction”, 2010.
2. Reza Zafarani, Mohammed Ali Abbasi, Huan Liu. “Social Media Mining: An Introduction”. Cambridge University Press 2014.
3. Maksim Tsvetovat and Alexander Kouznetsov. “Social Network Analysis for Startups”. O’Reilly Media, 2011.

E-Resources:

1. https://www.classcentral.com/course/edx-social-network-analysis-sna-9134
2. https://www.coursera.org/learn/social-network-analysis